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Tire pressure monitoring system, tire pressure monitoring program and tire pressure monitoring method

Abstract

The tire pressure monitoring system repeatedly acquires the internal pressure value of the pneumatic tire and activates the timer when the internal pressure value falls below the warning threshold. The tire pressure monitoring system stops the output of the warning information associated with the warning threshold even if the internal pressure value falls below the warning threshold again until the timer expires. The tire pressure monitoring system newly activates the timer when the internal pressure value exceeds the warning threshold after falling below the warning threshold or the danger threshold in the period until the timer expires.

Inventors:	Takasawa; Yuya (Tokyo, JP), Jakircevic; Thierry (Tokyo, JP)
Applicant:	BRIDGESTONE CORPORATION (Tokyo, JP)
Family ID:	1000008819030
Assignee:	BRIDGESTONE CORPORATION (Tokyo, JP)
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References Cited

U.S. PATENT DOCUMENTS

Patent No.	Issued Date	Patentee Name	U.S. Cl.	CPC
5710539	12/1997	Iida	N/A	N/A
2002/0126005	12/2001	Hardman et al.	N/A	N/A
2006/0016110	12/2005	Nakatani et al.	N/A	N/A
2007/0144639	12/2006	Oh	N/A	N/A
2010/0182142	12/2009	Svedberg et al.	N/A	N/A
2011/0190975	12/2010	Hourlier et al.	N/A	N/A
2013/0009763	12/2012	Koga	N/A	N/A
2014/0172342	12/2013	Yu	702/104	B60C23/0471
2017/0129294	12/2016	Taki	N/A	N/A
2017/0305213	12/2016	Taki	N/A	N/A
2020/0031180	12/2019	Usami	N/A	N/A

FOREIGN PATENT DOCUMENTS

Patent No.	Application Date	Country	CPC
102862449	12/2012	CN	N/A
103547464	12/2013	CN	N/A
110366498	12/2018	CN	N/A
102015115917	12/2016	DE	B60C23/0454
2 263 889	12/2013	EP	N/A
07-186643	12/1994	JP	N/A
2005-88613	12/2004	JP	N/A
2005-321315	12/2004	JP	N/A
2008-155795	12/2007	JP	N/A
2011-526560	12/2010	JP	N/A
2017-087957	12/2016	JP	N/A
2017-171240	12/2016	JP	N/A
2017-194412	12/2016	JP	N/A

OTHER PUBLICATIONS

Extended European Search Report dated Aug. 28, 2024 in Application No. 22787825.3. cited by applicant

International Search Report for PCT/JP2022/004725 dated Apr. 5, 2022 (PCT/ISA/210). cited by applicant

Office Action issued Jul. 9, 2025 in Chinese Patent Application No. 202280027576.X. cited by applicant

Primary Examiner: Vorce; Amelia

Attorney, Agent or Firm: Sughrue Mion, PLLC

Background/Summary**CROSS REFERENCE TO RELATED APPLICATIONS**

(1) This application is a National Stage of International Application No. PCT/JP2022/004725 filed Feb. 7, 2022, claiming priority based on Japanese Patent Application No. 2021-067132 filed Apr. 12, 2021.

TECHNICAL FIELD

(2) The present disclosure relates to a tire pressure monitoring system, a tire pressure monitoring program and a tire pressure monitoring method for monitoring the tire pressure mounted on a vehicle.

BACKGROUND ART

(3) Conventionally, a tire pressure monitoring system for monitoring the air pressure (internal pressure) of a pneumatic tire (hereinafter abbreviated as tire when appropriate) mounted on a vehicle has been widely used.

(4) The internal pressure value of the pneumatic tire measured by the sensor can vary depending on the temperature of the air chamber in the pneumatic tire mounted on the rim wheel and measurement error of the sensor. Therefore, a technique for controlling the occurrence timing of an alarm of an internal pressure drop using a counter for the measurement times is known (Patent Literature 1).

CITATION LIST

Patent Literature

(5) [Patent Literature 1]

(6) Japanese Patent Laid-Open Publication No. H7-186643

SUMMARY OF INVENTION

(7) However, when the internal pressure value fluctuates around the threshold value, which is the criterion of the alarm, the occurrence and stop of the alarm frequently occur, which places a large load on the tire pressure monitoring system, and there is also a concern that the user may feel annoyed.

(8) Therefore, the following disclosure has been made in view of this situation, and the purpose of the disclosure is to provide a tire pressure monitoring system, a tire pressure monitoring method, and a tire pressure monitoring program which can output the alarm of the internal pressure drop at an appropriate timing while preventing the occurrence and stop of the alarm frequently.

(9) One aspect of the present disclosure is a tire air pressure monitoring system (tire pressure monitoring system **10**) including an acquisition unit (tire data acquisition unit **110**) that repeatedly

acquires an internal pressure value of a pneumatic tire (tire **30**), a timer control unit (timer control unit **130**) that activates a timer when the internal pressure value becomes equal to or less than a warning threshold, and an alarm unit (alarm unit **150**) that outputs warning information until the timer expires when the internal pressure value becomes equal to or less than the warning threshold, and that outputs danger information until the timer expires when the internal pressure value becomes a danger threshold that is lower than the warning threshold. The timer control unit newly activates the timer when the internal pressure value exceeds the warning threshold value or the danger threshold value after falling below the warning threshold value or the danger threshold value in a period until the timer expires.

(10) One aspect of the present disclosure is a tire pressure monitoring program causing a computer to execute an acquisition process for acquiring an internal pressure value of a pneumatic tire, a timer control process for activating a timer when the internal pressure value becomes equal to or less than a warning threshold, and an alarming process for outputting warning information until the timer expires when the internal pressure value falls below the warning threshold value, and outputting danger information until the timer expires when the internal pressure value becomes equal to or less than the warning threshold, and that outputs danger information until the timer expires when the internal pressure value becomes a danger threshold that lower than the warning threshold. In the timer control process, newly activates the timer when the internal pressure value exceeds the warning threshold value or the danger threshold value after falling below the warning threshold value or the danger threshold value in a period until the timer expires.

(11) One aspect of the present disclosure is a tire pressure monitoring method including the steps of an acquisition step for repeatedly acquiring the internal pressure value of a pneumatic tire, a activating step for activating a timer when the internal pressure value becomes equal to or less than a warning threshold, and an outputting step for outputting warning information until the timer expires when the internal pressure value becomes equal to or less than the warning threshold, and that outputs danger information until the timer expires when the internal pressure value becomes a danger threshold that is lower than the warning threshold. In the activating step, newly activates the timer when the internal pressure value exceeds the warning threshold value or the danger threshold value after falling below the warning threshold value or the danger threshold value in a period until the timer expires.

Description

BRIEF DESCRIPTION OF DRAWINGS

- (1) FIG. **1** is a schematic diagram of the tire pressure monitoring system **10**.
- (2) FIG. **2** is a schematic plan view of the vehicle **20**.
- (3) FIG. **3** is a functional block diagram of the sensor unit **40**.
- (4) FIG. **4** is a functional block diagram of the tire condition monitoring server **100**.
- (5) FIG. **5** is a diagram showing a schematic operation flow of an alarm output of an internal pressure value by the tire condition monitoring server **100**.
- (6) FIG. **6** is a diagram showing an operation example of an alarm output of a conventional internal pressure value.
- (7) FIG. **7** is a diagram showing an operation example of an alarm output of an internal pressure value according to the present embodiment.

MODES FOR CARRYING OUT THE INVENTION

(8) Hereinafter, the embodiment will be described based on the drawings. The same functions and structures are denoted by the same or similar reference numerals, and the description thereof will be omitted accordingly.

(1) Overall Outline of Tire Pressure Monitoring System

- (9) FIG. 1 is an overall outline of the tire pressure monitoring system **10** according to the present embodiment. As shown in FIG. 1, the tire pressure monitoring system **10** provides monitoring services for a fleet including a plurality of vehicles **20**, such as trucks. The tire pressure monitoring system **10** essentially monitors the status of individual vehicles **20** at a fixed point, such as within a base (yard) of a user **80** where the plurality of vehicles **20** are parked.
- (10) In this embodiment, the tire pressure monitoring system **10** monitors the status of the vehicles **20**, specifically, the pneumatic tires mounted on the vehicles **20**, specifically, the status of tires **30** (Not shown in FIG. 1, see FIG. 2).
- (11) The tire pressure monitoring system **10** includes a tire condition monitoring server **100** that monitors the status of the plurality of tires **30** mounted on the vehicle **20** at a location separated from the vehicles **20**, for example, in the yard of the user **80**, using a plurality of sensors associated with each of the plurality of tires **30**.
- (12) In this embodiment, the vehicle **20** is a vehicle used for business in a company, such as a truck. Therefore, the vehicle **20** may be other than a truck, for example, a bus or a taxi. The plurality of vehicles **20** are operated by the user **80** such as a company. However, the user **80** does not necessarily have to be a company (business entity) and may be a non-profit-making organization or individual.
- (13) The tire condition monitoring server **100** is connected to a communication network **70**. The communication network **70** is a communication network using a wired or wireless communication system and may include the Internet.
- (14) The communication network **70** also includes a wireless access network called a so-called low-power wide area network (LPWA: Low Power Wide Area or LPWAN: Low Power Wide Area Network, hereinafter abbreviated as LPWA when appropriate).
- (15) LPWA is a wireless access system having coverage that cannot be satisfied by short-range wireless systems such as Bluetooth (registered trademark), and is characterized by low power consumption, low bit rate, and wide coverage.
- (16) Examples of services using LPWA include Sigfox (registered trademark), LoRa (registered trademark) and Wi-Fi HaLow. In this embodiment, the tire pressure monitoring system **10** assumes the use of services using LPWA. However, LPWA may be called by a different name, provided that it is a wireless communication system that has a wider coverage than a near-field communication system such as Bluetooth® and achieves lower power consumption than a mobile communication system such as 4G/LTE (Long Term Evolution).
- (17) A base station **60** is a wireless base station in accordance with LPWA and is a component of a low-power wide area network included in the communication network **70**. A plurality of base stations **60** are arranged in a geographically distributed manner in order to secure a communication range by the low-power wide-area network.
- (2) Vehicle Configuration
- (18) FIG. 2 is a schematic plan view of the vehicle **20**. As described above, in this embodiment, the vehicle **20** is assumed to be a truck.
- (19) As shown in FIG. 2, the vehicle **20** has a front wheel axle **21** and a rear wheel axle **22**. The front wheel axle **21** is a steering shaft and is located on the front side of the vehicle **20**. The rear wheel axle **22** is a driving shaft and is located on the rear side of the vehicle **20**, that is, on the loading compartment (loading platform) side. The rear wheel axle **22** is a so-called double tire, and two tires **30** are mounted on each side.
- (20) A total of 6 tires **30** are mounted on the vehicle **20** having such an axle configuration. Each tire **30** is mounted with a sensor unit **40**. Specifically, the sensor unit **40** is mounted on the inner surface of a rim wheel (not shown) to which the tire **30** is assembled. The mounting position of the sensor unit **40** may not necessarily be the inner surface (well portion) of the rim wheel, may be the inner surface of the tire **30**, or may be the base side of the air valve of the rim wheel.
- (21) The sensor unit **40** senses the state of the tire **30**. Specifically, the sensor unit **40** measures the

air pressure (internal pressure) and temperature of the tire **30** assembled to the rim wheel. In addition to the internal pressure and temperature, the sensor unit **40** may measure, for example, distortion.

(22) The sensor unit **40** has a wireless communication function, specifically, a wireless communication function in accordance with LPWA, and can perform wireless communication with the base station **60**.

(3) Functional Block Configuration of the Sensor Unit and Tire Pressure Monitoring System

(23) Next, the functional block configuration of the sensor unit **40** and the tire condition monitoring server **100** will be described.

(3.1) Sensor Unit **40**

(24) FIG. **3** is a functional block diagram of the sensor unit **40**. As shown in FIG. **3**, the sensor unit **40** includes a temperature sensor **41**, a pressure sensor **43**, an acceleration sensor **44**, a sensor ID setting unit **45**, a wireless communication unit **47**, and a battery **49**.

(25) The temperature sensor **41** measures the temperature in the air chamber of the tire **30** attached to the rim wheel.

(26) For example, a semiconductor type can be used as the temperature sensor **41**.

(27) The pressure sensor **43** measures the pressure (internal pressure) in the air chamber of the tire **30**. As the pressure sensor **43**, for example, a capacitance type can be used.

(28) The acceleration sensor **44** detects the acceleration of the tire **30** in a predetermined direction. In this embodiment, the acceleration sensor **44** can detect the acceleration of the tire **30** in tire radial direction, specifically, the centrifugal acceleration (which may be rephrased as centrifugal force). In addition to tire radial direction, the acceleration sensor **44** may also detect the acceleration in the tire width direction. As the acceleration sensor **44**, a general-purpose acceleration sensor such as a 3-axis acceleration sensor may be used.

(29) The sensor ID setting unit **45** sets a sensor ID (sensor identification information) for identifying the sensor unit **40**. Specifically, the sensor ID setting unit **45** stores the identification information of the sensor unit **40** and provides the stored identification information to the wireless communication unit **47**. Further, more information (e.g., tire type) may be included. The sensor identification information is multiplexed on a radio signal transmitted from the radio communication unit **47**.

(30) The radio communication unit **47** has a radio communication function in accordance with LPWA. Specifically, the radio communication unit **47** transmits to the base station **60** a radio signal (radio wave) in which data indicating the measured temperature output from the temperature sensor **41** and data indicating the measured pressure (internal pressure) output from the pressure sensor **43** are multiplexed.

(31) The temperature sensor **41** and the pressure sensor **43** may not measure the temperature and internal pressure during the period when the sensor unit **40** detects acceleration, that is, during the period when the tire **30** is rolling.

(32) The battery **49** supplies the necessary power to each functional block constituting the sensor unit **40**. Specifically, although the type of the battery is not particularly limited, it is preferable that the battery **49** be composed of a primary battery or the like capable of continuously driving the sensor unit **40** over a long period of time (For example, one year or more).

(3.2) Tire Condition Monitoring Server **100**

(33) FIG. **4** is a functional block configuration diagram of the tire condition monitoring server **100**. As shown in FIG. **4**, the tire condition monitoring server **100** includes a tire data acquisition unit **110**, a timer control unit **130**, and an alarm unit **150**.

(34) These function blocks are realized by executing a computer program (software) on hardware such as a server computer.

(35) Specifically, the tire condition monitoring server **100** includes, as hardware elements, a processor, a memory, an input device, a display, and an external interface. The computer program

(software) may also be provided via the communication network **70** or recorded on a computer-readable recording medium such as an optical disk, a hard disk drive, or a flash memory.

(36) The tire data acquisition unit **110** acquires tire data indicating the state of the tire **30** from the sensor unit **40** (see FIGS. 2 and 3). Specifically, the tire data acquisition unit **110** directly acquires tire data via wireless communication from a plurality of sensors such as the temperature sensor **41** and the pressure sensor **43** that are associated with the plurality of tires **30**, that is, constitute the sensor unit **40** attached to the tires **30** (Specifically, rim wheels, hereinafter the same).

(37) The tire data include at least internal pressure data measured by the pressure sensor **43**. The internal pressure data may be data from which the internal pressure value of the tire **30** can be determined. The internal pressure value may be indicated directly, or it may be an index with a smaller number of bits than that indicated directly.

(38) The tire data may also include temperature data measured by the temperature sensor **41**. In addition, if the sensor unit **40** includes other sensors (For example, an acceleration sensor), the tire data may include data measured by the sensors.

(39) The tire data acquisition unit **110** repeatedly acquires tire data from the sensor unit **40** through the base station **60** and the communication network **70**. In this embodiment, the tire data acquisition unit **110** constitutes an acquisition unit for repeatedly acquiring the internal pressure value of the pneumatic tire.

(40) In this embodiment, the tire data acquisition unit **110** acquires tire data via wireless communication via a low-power wide-area network (LPWA). That is, the tire data acquisition unit **110** directly acquires tire data via wireless communication using LPWA from the sensor unit **40**, rather than via the vehicle **20** using a short-range wireless method from the sensor unit **40**.

(41) The tire data acquisition timing (repetition period) is not particularly limited, but is generally set to about 1 to 10 minutes, for example. In consideration of the characteristics of LPWA, the acquisition timing (repetition period) may be longer.

(42) The timer control unit **130** controls a timer used for determining the timing for outputting information indicating abnormality of tire data. Specifically, the timer control unit **130** can control a timer used for determining the timing for outputting warning information and danger information.

(43) Cautionary information may be referred to as Warning. Warning may be interpreted as information alerting the user **80** (and/or driver of vehicle **20**, hereinafter the same) to a decrease in internal pressure. Warning may be positioned as an alert alerting the user **80** to a decrease in internal pressure but requiring no immediate action.

(44) The danger information may be referred to as Critical. Critical may be interpreted as information that requests the user **80** to respond immediately, for example, to stop the vehicle **20** from running and check the tire condition.

(45) The warning information may be associated with a threshold of the internal pressure value (called the warning threshold), and the danger information may be associated with a threshold of the internal pressure value lower than the warning threshold (called the danger threshold).

(46) For example, the warning threshold may be a 5~10% reduction in the recommended internal pressure value, and the danger threshold may be a 20% reduction in the recommended internal pressure value. The recommended internal pressure value may be the internal pressure value specified by the manufacturer of the vehicle **20**, or the air pressure corresponding to the maximum load capacity in the JATMA (Japan Automobile Tire Manufacturers Association) YearBook (may be referred to as normal internal pressure). It should be noted that ETRTO in Europe, TRA in the United States, and other national tire standards are applicable.

(47) When the internal pressure value becomes equal to or less than the warning threshold, the timer control unit **130** starts the timer. Specifically, when the internal pressure value based on the tire data acquired by the tire data acquisition unit **110** becomes equal to or less than the warning threshold, the timer control unit **130** may activate (start) the timer.

(48) On the other hand, when the internal pressure value acquired by the tire data acquisition unit

110 becomes equal to or less than the warning threshold or the danger threshold, and then exceeds the warning threshold, the timer control unit **130** may activate a new timer. Even if the internal pressure value becomes equal to or less than the warning threshold or the danger threshold before the expiration of the newly started timer, the timer control unit **130** need not restart the timer.

(49) Further, the timer control unit **130** may continue the timer without restarting the timer when the internal pressure value becomes higher than the critical threshold value and becomes lower than the warning threshold value after the internal pressure value becomes lower than the critical threshold value. That is, the timer control unit **130** may continue the timer without stopping and restarting (resetting) the timer when the internal pressure value below the critical threshold value exceeds the critical threshold value but is lower than the warning threshold value.

(50) The alarm unit **150** executes processing related to the alarm of the internal pressure value based on the warning threshold value or the warning threshold value. Specifically, the alarm unit **150** can output warning information (Warning) associated with the warning threshold value or danger information (Critical) associated with the warning threshold value.

(51) The alarm unit **150** can notify the warning information or danger information to a predetermined destination associated with the user **80**. The predetermined destination may be an e-mail address of the driver of the user **80** or the vehicle **20**, or it may be an ECU (Electronic Control Unit) installed in the vehicle **20**.

(52) The alarm unit **150** outputs warning information when the internal pressure value acquired by the tire data acquisition unit **110** is equal to or less than the warning threshold. The alarm unit **150** outputs danger information when the internal pressure value is equal to or less than the danger threshold.

(53) Specifically, when the internal pressure value becomes equal to or less than the warning threshold, the alarm unit **150** may output warning information until the timer activated by the timer control unit **130** expires. When the internal pressure value becomes the danger threshold, the alarm unit **150** may output danger information until the timer expires. That is, the alarm unit **150** may continue to output warning information or danger information until the timer expires.

(54) In addition, the alarm unit **150** may not output warning information when the internal pressure value acquired by the tire data acquisition unit **110** becomes higher than the danger threshold and less than the warning threshold after it becomes less than the danger threshold. In this case, as described above, the timer activated (restarted) by falling below the danger threshold may be continued without being reset.

(4) Operation of the Tire Pressure Monitoring System

(55) Next, the operation of the tire pressure monitoring system **10** will be described. Specifically, the operation related to the output of warning information and danger information (critical) of the internal pressure value by the tire condition monitoring server **100** will be described.

(4.1) Schematic Operation Flow

(56) FIG. 5 shows a schematic operation flow of an alarm output of an internal pressure value by the tire condition monitoring server **100**. As shown in FIG. 5, the tire condition monitoring server **100** acquires an internal pressure value of the tire **30** (S10).

(57) Specifically, the tire condition monitoring server **100** receives the internal pressure value based on the tire data transmitted from the sensor unit **40**.

(58) The tire condition monitoring server **100** executes a Warning/Critical determination based on the acquired internal pressure value (S20). Specifically, the tire condition monitoring server **100** determines whether the acquired internal pressure value is lower than the warning threshold or the danger threshold. As described above, the warning threshold can be a 5~10% reduction value of the recommended internal pressure value, and the danger threshold can be a 20% reduction value of the recommended internal pressure value. In the following example, the warning threshold value is a 10% reduction value of the recommended internal pressure value, and the danger threshold value is a 20% reduction value of the recommended internal pressure value.

(59) The tire condition monitoring server **100** starts or restarts the timer based on the Warning/Critical determination result of the internal pressure value (**S30**). For example, the tire condition monitoring server **100** activates (starts) the timer when the internal pressure value becomes equal to or less than the warning threshold. The tire condition monitoring server **100** may restart the timer when the internal pressure value becomes equal to or less than the danger threshold value which is lower than the warning threshold value in the period until the activated timer expires.

(60) Further, the tire condition monitoring server **100** may activate the new timer when the internal pressure value becomes equal to or less than the warning threshold value or danger threshold value and then exceeds the warning threshold value, that is, when the internal pressure value returns normally.

(61) The restart of the timer may mean that the timer started when the internal pressure value falls below the warning threshold is restarted when the internal pressure value falls below the danger threshold. In a broad sense, when the internal pressure value is lower than the warning threshold, the timer is restarted when the severity changes, specifically, when the severity changes from below the warning threshold to below the danger threshold, or when the severity changes from below the danger threshold to above the danger threshold to below the warning threshold.

(62) On the other hand, activating the new timer may mean starting the timer once the internal pressure value that is below the warning threshold or danger threshold exceeds the warning threshold.

(63) The tire condition monitoring server **100** outputs an alarm according to the internal pressure value (**S40**). Specifically, the tire condition monitoring server **100** outputs warning information or danger information according to the internal pressure value.

(64) The time from the start to the expiration of the timer may be set appropriately according to the type of the vehicle **20** and/or the tire **30**, the operating environment, etc.

(4.2) Example

(65) A specific operation example of the alarm output of the internal pressure value by the tire condition monitoring server **100** will be described below.

(66) FIG. **6** shows an operation example of the alarm output of the conventional internal pressure value. In this operation example, the recommended internal pressure value (prescribed internal pressure) is 120 psi (pound per square inch). In this operation example, the warning threshold value (Warning) is set to a 10% reduction value (108 psi) of the recommended internal pressure value, and the danger threshold value (Critical) is set to a 20% reduction value (96 psi) of the recommended internal pressure value. In addition, the internal pressure value may be acquired repeatedly in a 5-minute cycle, for example.

(67) Note that bar or kPa may be used as the unit, rather than psi. A catastrophic threshold associated with an internal pressure value lower than the danger threshold may also be obtained. These internal pressure values may be set by the user **80** as appropriate.

(68) As shown in FIG. **6**, when it is determined whether the acquired internal pressure value has fallen below the warning threshold or the danger threshold without applying a timer and the warning information or the danger information is output, the occurrence and stop of the warning information or the danger information may be repeated frequently.

(69) Such an event is caused mainly because the internal pressure value fluctuates up and down around the threshold due to at least one of the accuracy (measurement error) of the sensor unit **40** (pressure sensor **43**) and the fluctuation of the temperature of the air chamber in the tire **30**.

(70) In the operation of outputting the warning information or the danger information (alarm) in accordance with the acquired internal pressure value as it is, the occurrence and stopping of the warning information or the danger information are likely to cause a problem.

(71) Specifically, the occurrence and stopping of the alarm occur frequently, and a large load is applied to the tire condition monitoring server **100**. If the occurrence and stopping of the alarm

occur frequently, the user **80** may feel annoyed. It should be noted that even when the counter disclosed in Patent Literature 1 is used, it is considered that it is not sufficient to solve the above-mentioned problem when the occurrence and stopping of the alarm occur frequently.

(72) FIG. 7 shows an operation example of the alarm output of the internal pressure value according to the present embodiment. In order to more reliably solve the problem described above, a time limit on the output of warning information and danger information (alarm) concerning the internal pressure value is applied.

(73) Specifically, even if an event corresponding to warning information or danger information, that is, the internal pressure value becomes less than or equal to the warning threshold or the internal pressure value becomes less than or equal to the danger threshold, the event may be ignored within a predetermined time period, that is, until the timer setting time has elapsed.

(74) As described above, the timer may be started (or restarted) whenever an event corresponding to warning information or danger information (which may be called an open issue) occurs (create). If the event changes from Warning to Critical, that is, the severity of the event increases, the timer may be restarted.

(75) Note that the issue may refer to an event (an identification symbol for internal processing) from the occurrence of the above-mentioned failure event related to one air pressure until its resolution.

(76) The setting time of the restarted timer may be different from that of the timer started at the timing below the warning threshold (For example, a shorter time). The setting time of the timer started at the time below the danger threshold may be different from that of the timer started at the time below the warning threshold.

(77) On the other hand, when the internal pressure value returns from the danger information to the level of the warning information, that is, when the internal pressure value below the danger threshold is above the danger threshold but below the warning threshold, the tire condition monitoring server **100** may ignore the change of such event and continue without stopping or restarting (resetting) the timer.

(78) When the internal pressure value becomes below the warning threshold or the danger threshold and then exceeds the warning threshold (That is, if the pressure drop problem is resolved,), the timer may be started anew. In this case, the setting time of the timer may be the same as the timer started when the internal pressure value becomes below the warning threshold in the past, but it may be changed.

(79) For this reason, events corresponding to warning or danger information that occurred from the time the problem of the internal pressure drop was resolved until the timer expired may be ignored (including a change from Warning to Critical).

(80) Thus, the tire condition monitoring server **100** may activate (start) a timer (which may be called the first timer) when an event corresponding to the warning or danger information occurs, and may start another timer (which may be called the second timer) when an event corresponding to the warning information disappears (Close).

(81) The setting time of the timer may be dynamically changed according to the number of times the event occurs or stops (the number of repetitions). For example, if the number of repetitions exceeds the predetermined number (the first predetermined number), the setting time of the timer may be extended. Conversely, if the number of repetitions falls below the predetermined number (the second predetermined number), the setting time of the timer may be shortened.

(82) In addition, when the danger information is output while the warning information is output, the warning information may be stopped. Furthermore, when an internal pressure value corresponding to the warning information or the danger information is acquired consecutively for a predetermined number of times, the alarm corresponding to the warning information or the danger information may be output regardless of whether the timer has expired. On the other hand, when a normal internal pressure value exceeding the warning threshold is obtained for a predetermined

number of consecutive times, the alarm corresponding to the warning information or the danger information may be stopped regardless of whether the timer has expired.

(5) Operational Effects

(83) According to the above-described embodiment, the following operational effects can be obtained. Specifically, the tire pressure monitoring system **10** can start a new timer when the internal pressure value exceeds the warning threshold value after falling below the warning threshold value or the danger threshold value in the period until the timer expires.

(84) Therefore, when it is determined that the internal pressure value of the tire **30** has returned to normal, the output of the warning information or the danger information can be controlled based on the setting time of the new timer, and the warning information or the danger information can be notified at an appropriate timing and period.

(85) In particular, even when the internal pressure value fluctuates around the threshold value which is the standard of the alarm, the occurrence and stopping of the alarm (the warning information or the danger information) frequently occurs, and the risk of the occurrence of the so-called “false positive” alarm can be reduced. As a result, the load of the tire pressure monitoring system **10** can be reduced, and the user can avoid feeling annoyance. That is, according to the tire pressure monitoring system **10**, an alarm of an internal pressure drop can be outputted at an appropriate timing while preventing frequent occurrence of an alarm and a stop.

(86) In the present embodiment, the tire pressure monitoring system **10** does not restart the timer even if the internal pressure value becomes less than the warning threshold or the danger threshold before the newly started timer expires when the warning threshold is exceeded. Therefore, the warning information or the danger information can be notified at a more appropriate timing and period.

(87) In this embodiment, the tire pressure monitoring system **10** may continue the timer without outputting the warning information when the internal pressure value becomes higher than the danger threshold and less than the warning threshold after the internal pressure value becomes less than or equal to the danger threshold. Therefore, it is possible to avoid the output of an alarm that requires less notification, further reduce the load of the tire pressure monitoring system **10**, and more reliably avoid that the user feels annoyance.

(6) Other Embodiments

(88) Although the embodiments have been described above, they are not limited to the description of the embodiments, and it is obvious to those skilled in the art that various modifications and improvements can be made.

(89) For example, in the above-described embodiment, the tire condition monitoring server **100** is connected to the communication network **70** and implemented by executing a computer program (software) on hardware such as a server computer, but some or all of the functions of the tire condition monitoring server **100** may be provided virtually by a combination of services provided on a network cloud (not shown).

(90) In addition, in the above-described embodiment, the sensor unit **40** has a function of performing wireless communication with the base station **60** according to LPWA, but may further have a function of performing wireless communication with the vehicle **20** using a short-range wireless method as well as a conventional method.

(91) Although embodiments of the present invention have been described as described above, the discussion and drawings that form part of this disclosure should not be understood as limiting the present invention. The present disclosure will reveal to those skilled in the art a variety of alternative embodiments, embodiments and operational techniques.

EXPLANATION OF REFERENCE NUMERALS

(92) **10** tire pressure monitoring system **20** vehicle **21** front wheel axles **22** rear wheel axles **30** tire **40** sensor unit **41** temperature sensor **43** pressure sensor **44** acceleration sensor **45** sensor ID setting unit **47** wireless communication unit **49** battery **60** base station **70** communication network **80** user

Claims

1. A tire air pressure monitoring system comprising: an acquisition unit that repeatedly acquires an internal pressure value of a pneumatic tire; a timer control unit that activates a timer when the internal pressure value becomes equal to or less than a warning threshold; and an alarm unit that outputs warning information until the timer expires when the internal pressure value becomes equal to or less than the warning threshold, and that outputs danger information until the timer expires when the internal pressure value becomes a danger threshold that is lower than the warning threshold, wherein the timer control unit newly activates the timer when the internal pressure value exceeds the warning threshold value or the danger threshold value after falling below the warning threshold value or the danger threshold value in a period until the timer expires.
 2. The tire air pressure monitoring system according to claim 1, wherein the timer control unit does not re-activate the timer even when the internal pressure value falls below the warning threshold value or the danger threshold value before the newly activated timer expires.
 3. A tire pressure monitoring program causing a computer to execute: an acquisition process for acquiring an internal pressure value of a pneumatic tire; a timer control process for activating a timer when the internal pressure value becomes equal to or less than a warning threshold; and an alarming process for outputting warning information until the timer expires when the internal pressure value falls below the warning threshold value, and outputting danger information until the timer expires when the internal pressure value becomes equal to or less than the warning threshold, and that outputs danger information until the timer expires when the internal pressure value becomes a danger threshold that is lower than the warning threshold; wherein in the timer control process, newly activates the timer when the internal pressure value exceeds the warning threshold value or the danger threshold value after falling below the warning threshold value or the danger threshold value in a period until the timer expires.
 4. A tire pressure monitoring method comprising the steps of: an acquisition step for repeatedly acquiring the internal pressure value of a pneumatic tire; an activating step for activating a timer when the internal pressure value becomes equal to or less than a warning threshold; and an outputting step for outputting warning information until the timer expires when the internal pressure value becomes equal to or less than the warning threshold, and that outputs danger information until the timer expires when the internal pressure value becomes a danger threshold that is lower than the warning threshold, wherein in the activating step, newly activates the timer when the internal pressure value exceeds the warning threshold value or the danger threshold value after falling below the warning threshold value or the danger threshold value in a period until the timer expires.
 5. The tire pressure monitoring system according to claim 2, wherein when the internal pressure value is higher than the danger threshold and falls below the warning threshold after the internal pressure value falls below the danger threshold value, the alarm unit does not output the warning information, and the timer control unit continues the currently running timer.
 6. The tire pressure monitoring system according to claim 1, wherein when the internal pressure value is higher than the danger threshold and falls below the warning threshold after the internal pressure value falls below the danger threshold value, the alarm unit does not output the warning information and the timer control unit continues the currently running timer.
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